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Studierichting: Informatica
Oefeningenreeks 4A nr. 44.12

$$\begin{aligned} & \int \frac{9x^2 + x + 4}{x^3 - x^2 + x - 1} dx \\ &= \int \frac{9x^2 + x + 4}{(x-1)(x^2+1)} dx \\ &\Rightarrow \frac{9x^2 + x + 4}{(x-1)(x^2+1)} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} \end{aligned}$$

$x = 1$:

$$C = \frac{9(1)^2 + 1 + 4}{1^2 + 1} = \frac{14}{2} = 7$$

$x = i$:

$$\begin{aligned} Ai + B &= \frac{9(i)^2 + i + 4}{i - 1} = \frac{-5 + i}{-1 + i} \\ &= \frac{(-5 + i)(-1 - i)}{(-1 + i)(-1 - i)} = \frac{5 + 5i - i + 1}{2} = \frac{6 + 4i}{2} = 3 + 2i \\ &\Rightarrow A = 2 \text{ en } B = 3. \end{aligned}$$

$$\begin{aligned} &\Rightarrow \int \left(\frac{2x+3}{x^2+1} + \frac{7}{x-1} \right) dx \\ &= \int \frac{2x}{x^2+1} dx + 3 \int \frac{1}{x^2+1} dx + 7 \int \frac{1}{x-1} dx \\ &= \ln(x^2+1) + 3 \operatorname{Bgtan}(x) + 7 \ln|x-1| + C \end{aligned}$$

References